



**Bachelor's Degree** Programme,  
2nd Semester of the 3rd Year  
Course Module: **Cloud Computing** Lecturer:  
Osvaldo Santos

## Practical assignment on virtual networks in Azure

### Purpose of this document

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Introduction to the tools for creating and configuring virtual networks on the Azure public cloud computing platform

### Bibliography and useful links

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Azure Virtual Networks overview

<https://learn.microsoft.com/en-us/azure/virtual-network/virtual-networks-overview>

Azure Virtual Networks and Subnets

<https://learn.microsoft.com/en-us/azure/virtual-network/network-overview>

Azure Command-Line Interface (CLI) documentation

<https://learn.microsoft.com/en-us/cli/azure/>



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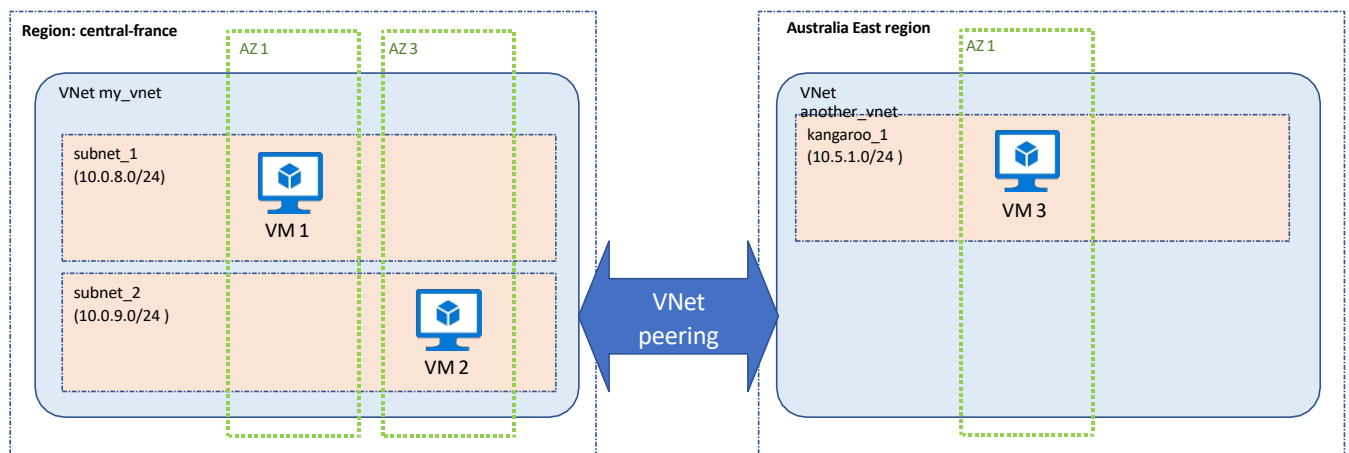
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In this assignment, you will use your ipcbcampus student account to access Azure

**At the end of the assignment, you must delete all Azure resources so that you can preserve your balance**

The scenario to be set out in this lab sheet is as follows:



The diagram shows two VNets, each in a different region, interconnected via VNet peering

Each VNet contains one or more subnets.

### 1) Creating a virtual network via the portal

- Create a resource group called "class" in the Central France region.
- Create a VNet named "my\_vnet" in the Central France region, within the "class" resource group, with the following characteristics:
  - Address space 10.0.0.0/16
  - Two subnets: subnet\_1: 10.0.8.0/24 and subnet\_2: 10.0.9.0/24, without a NAT gateway.
  - No bastion host, no DDoS protection, no firewall



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## Practical assignment on virtual networks in Azure

### 2) Creating a VM in this region and in this VNet

a) In the 'Class' resource group, in the 'Central France' region, create a VM with Apache to set up a web server with the following specifications:

#### i) Basics

- Name: vm1
- Availability options: Availability zone
- Availability zone where the VM will be created. Zone 1
- Security type: Standard
- Image: Ubuntu Server 20.04 LTS -x64 Gen2
- Size: Standard\_B1ms: 1 vCPU, 2 GB RAM
- Password authentication: username: student, password: Student1234567!
- Open inbound ports: SSH, HTTP, HTTPS

#### ii) Disks

- Standard SSD disk
- Do not change the other options

#### iii) Networking

- Virtual network: my\_vnet
- Subnet: subnet\_1
- Public IP address: (new) vm1-ip
- Do not change the other options

#### iv) Management

- Do not change anything



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## Practical assignment on virtual networks in Azure

### v) Advanced

- Place the cloud-init script to install Apache and create the home page with the server name. Note: you can copy this script from Moodle

```
#cloud-config
package_upgrade: true
packages:
  - apache2
runcmd:
  - echo "Server 1" > /var/www/html/index.html
  - systemctl restart apache2
```

- Do not change the other options

### vi) Tags

- Do not change anything

b) Create the VM and wait until it comes online.

c) What is the private IP address within subnet\_1? Answer: \_\_\_\_\_. Is it an IP address that is part of the subnet\_1 address space?

d) What is the VM's public IP address? Answer: \_\_\_\_\_. Test the web server using a browser and the VM's public IP address

e) Now, open an SSH terminal to the VM using the public IP address. Use the login credentials you set up when you created the VM.

f) In the SSH terminal, run the command "ip a", which displays the VM's network interfaces. Do two network interfaces appear (one with the private IP and the other with the public IP)? Or does only the interface with the private IP appear? Why? What is happening?

g) Still in the SSH terminal, try testing the VM's outbound access, for example by downloading a file from a web server using the command "wget <http://www.google.com>". Does the VM have outbound access to the Internet?



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## Practical assignment on virtual networks in Azure

### 3) Create another VM in the same region and the same VNet, but in a different availability zone and a different subnet

- a) Create another VM with the same specifications as vm1, except:
  - i) Name: vm2
  - ii) Availability zone where the VM will be created: Zone 3
  - iii) Subnet: subnet\_2
  - iv) No public IP address
  - v) Message on the web server: - echo "Server 2" > /var/www/html/index.html
- b) Create the VM vm2 and wait until it becomes active.
- c) What is the private IP address within subnet\_2? Answer: \_\_\_\_\_. Is it an IP address that is part of the subnet\_2 address space?
- d) Does any public IP address appear? Without a public IP, is it possible to access the vm2 web server?
- e) Without a public IP, is it possible to log in remotely via SSH to vm2? Are there other ways we can open a terminal for the VM?
- f) Now, in the Azure portal, open a serial console for vm2 (side menu, at the bottom – Serial Console). If prompted for authentication, use the login credentials you set up when you created the VM.
- g) In the console, run the command “ip a”, which displays the VM’s network interfaces. Does the private IP address match what the Azure portal showed?
- h) vm2 and vm1 are part of the same region, the same VNet, but are in different availability zones (different data centres) and on different subnets. Is there connectivity between these two VMs? What do you think?
- i) Now try pinging the private IP address of vm1. Is there connectivity between the two VMs? What is the average latency between the two VMs? Answer: \_\_\_\_\_. Is this a good or bad value?
- j) While still in the console, try testing the VM’s outbound access, for example by downloading a file from a web server using the command “wget <http://www.google.com>”. Does the VM have outbound access to the internet?



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## Practical assignment on virtual networks in Azure

### 4) Creating a VM in another region, on another VNet

- a) Create a new VNet called "outra\_vnet" in the "Australia East" region, in the "aula" resource group (it doesn't matter if the "aula" resource group is in France), with the following characteristics:
  - Address space 10.5.0.0/16
  - One subnet: kangaroo\_1: 10.5.1.0/24, without a NAT gateway.
  - No bastion host, no DDoS protection, no firewall
- b) Create another VM, with the same characteristics as vm1, except:
  - i) Name: vm3
  - ii) Region: Australia East
  - iii) Availability zone where the VM will be created. Zone 1
  - iv) VNet: outra\_vnet
  - v) Subnet: kangaroo\_1 Question: In the list of subnets, do subnet\_1 and subnet\_2 appear? Why?
  - vi) No public IP
  - vii) Message on the web server: - echo "Server 3" > /var/www/html/index.html
- c) Create the VM and wait until it comes online.
- d) What is the private IP address within the kanguru\_1 subnet? Answer: \_\_\_\_\_. Is it an IP address that forms part of the kanguru\_1 subnet address space?
- e) Now, open a console for vm3. Use the login credentials you set up when you created the VM.
- f) Try running a ping connectivity test from vm3 to vm1 and vm2. Is there connectivity? Why?

## Practical assignment on virtual networks in Azure

### 5) VNet peering

- a) In the VNet a\_minha\_vnet, create a Peering named “interconnection” to connect this VNet to the VNet outra\_vnet
- b) Try the connectivity from vm3 to vm1 and vm2 again. You should now be able to establish connectivity between VMs in different regions and different VNets.
- c) What is the average latency of communication between vm3 and vm1? Answer: \_\_\_\_\_. Compare this value with the latency value between vm1 and vm2, which you determined previously. What is the explanation for such a large difference?

### 6) Use of Bastion

- d) One way to access VM settings, even when there is no internet connectivity for those VMs, is to use a Bastion.
- e) In the “outra\_vnet” VNet, create a Bastion Host (there is an option for this in the side menu). It may take around 10 minutes to become available.
- f) Now, use the bastion to connect to vm3
- g) What is the average latency of the communication between vm3 and vm1? Answer: \_\_\_\_\_. Compare this value with the latency value between vm1 and vm2, which you determined previously. What is the explanation for such a large difference?

### 7) Create VNet, subnets and VMs using a Bash script

- h) Delete the resource group in which you created all these resources
- i) Now, write a Bash script that creates a resource group, creates a VNet in a region with a specific address space, creates two subnets with specific address blocks, and finally creates two VMs, one in each subnet. **Tip: don't try to write the whole script in one go; take it one step at a time and test each step. Start by writing the command to create the resource group.**
- j) Test the script using Cloud Shell.



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## Practical assignment on virtual networks in Azure

### 8) Conclusion

Now, delete all the resources